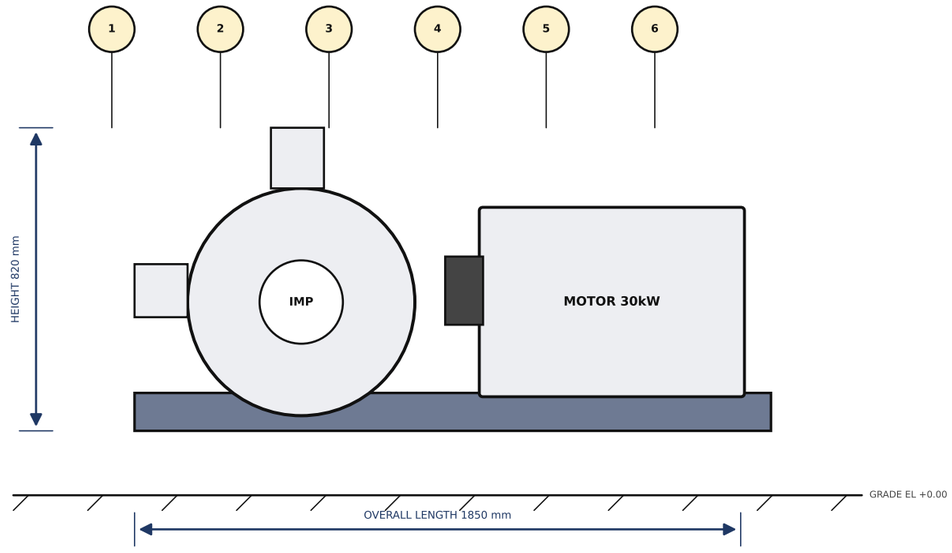
GENERAL ARRANGEMENT DRAWING - HEXANE FEED PUMP (GA-1201A)

Type: Centrifugal Pump (API 610 OH2) | Area 1200 - FEED PREPARATION & PURIFICATION



C/L ELEVATION 375 mm | DRY WEIGHT 690 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	SUCTION	3"-150#	RF	n-Hexane in
N2	DISCHARGE	2"-300#	RF	n-Hexane out
N3	SEAL FLUSH	1/2"	SW	Plan 11 flush
N4	CASING DRAIN	3/4"	SW	Drain
N5	CASING VENT	1/2"	SW	Vent

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	CASING	ASTM A351 CF8M	1
2	IMPELLER	ASTM A351 CF8M	1
3	SHAFT	AISI 4140	1
4	SHAFT SLEEVE	SS316	1
5	MECHANICAL SEAL	JC T2100	1
6	BEARING (DE)	7310 BECBM	1
7	BEARING (NDE)	6310 C3	1
8	BEARING HOUSING	CAST IRON	1
9	WEAR RING	SS316	2
10	COUPLING SPACER	AISI 4140	1
11	BASEPLATE	CARBON STEEL	1
12	OIL SEAL	NBR	2

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-1201, INTERLOCK SEQ-1201.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

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TITLE:
GENERAL ARRANGEMENT DRAWING

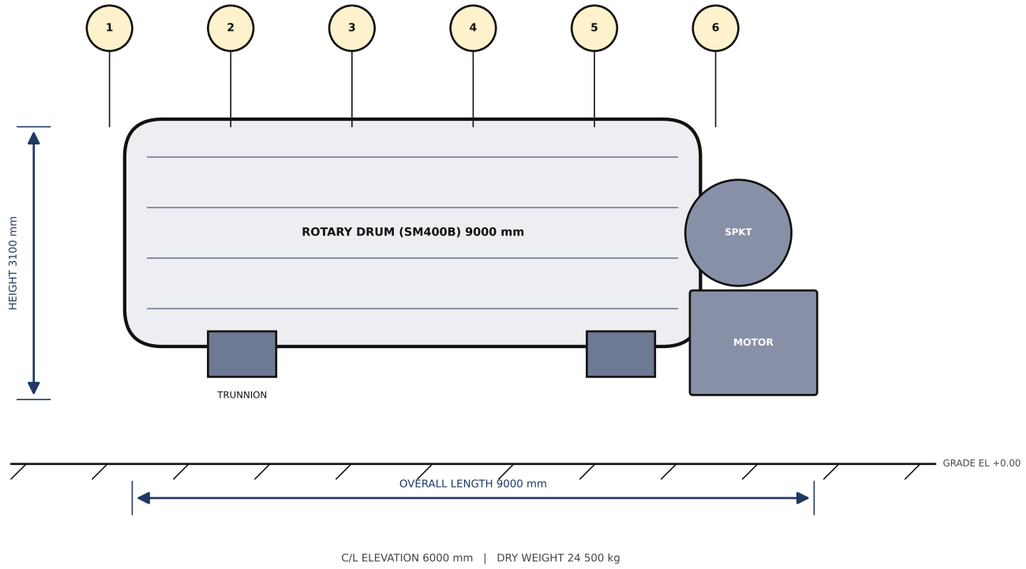
HEXANE FEED PUMP - Centrifugal Pump (API 610 OH2)

DWG No.
TJC-LLD-GA-GA-1201A

REV: 0
SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - POLYMER FLUID BED DRYER (YD-2301)

Type: Rotary Steam-Tube Dryer | Area 2300 - POLYMER SEPARATION & DRYING



C/L ELEVATION 6000 mm | DRY WEIGHT 24 500 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	POWDER INLET	12"	FLG	Wet powder from centrifuge
N2	POWDER OUTLET	12"	FLG	Dry powder to silo
N3	STEAM INLET	4"-150#	RF	LP steam via rotary joint
N4	CONDENSATE OUT	3"-150#	RF	Condensate return
N5	N2 PURGE	2"	RF	Nitrogen inerting
N6	VENT	6"	FLG	Vent to dust collector

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	BODY (DRUM)	SM400B	1
2	ROTOR SHAFT	SM400B/SS400	1
3	BEARING ASSEMBLY	-	2
4	SPROCKET	SCM440	1
5	ROTARY JOINT	KADANT ASSY	2
6	SUPPORT (TRUNNION)	SS400	2
7	GLAND PACKING	PTFE PILLER 4526L	8
8	GLAND PACKING	PTFE PILLER 4505L	2
9	OIL SEAL	NBR	2
10	OIL SEAL	NBR	2
11	O-RING	NBR	2
12	STEAM TUBE BUNDLE	SS304	1

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-2301, INTERLOCK SEQ-5500.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

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TITLE:
GENERAL ARRANGEMENT DRAWING

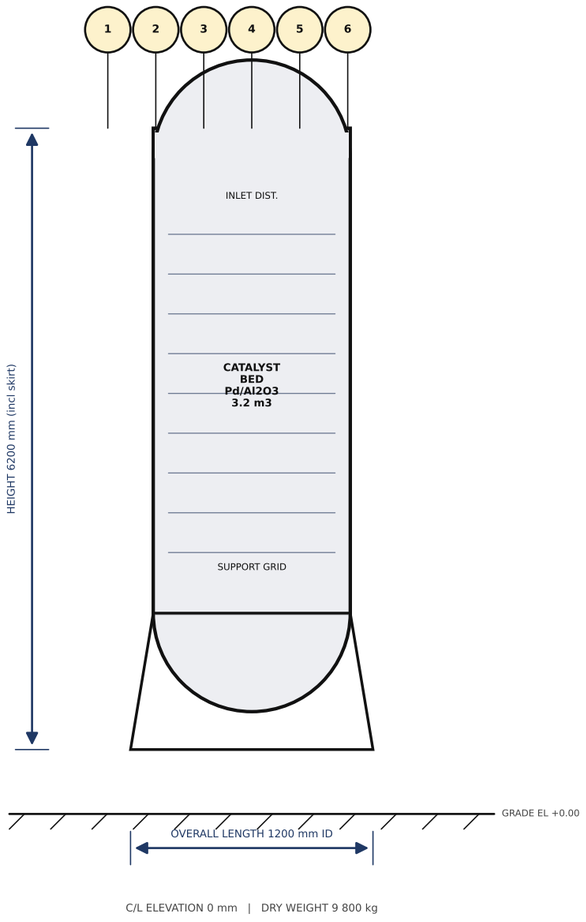
POLYMER FLUID BED DRYER - Rotary Steam-Tube Dryer

DWG No.
TJC-LLD-GA-YD-2301

REV: 0
SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - CATALYST REDUCTION REACTOR (DC-3401A)

Type: Fixed-Bed Reactor (Pd Catalyst) | Area 3400 - CATALYST PREPARATION



C/L ELEVATION 0 mm | DRY WEIGHT 9 800 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	INLET (TOP)	4"-150#	RF	N2/H2 in from heater
N2	OUTLET (BOTTOM)	4"-150#	RF	Gas out to analyzer
N3	PSV	3"-150#	RF	To PSV-3401
N4	TE MULTIPOINT	2"	FLG	TE-3401-1..8
M1	MANWAY TOP	24"	FLG	Catalyst load
M2	MANWAY BOTTOM	24"	FLG	Catalyst unload

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	SHELL	SA-516-70 + SS304 CLAD	1
2	TOP HEAD (2:1 ELLIP)	SA-516-70	1
3	BOTTOM HEAD	SA-516-70	1
4	CATALYST SUPPORT GRID	SS304	1
5	HOLD-DOWN GRID	SS304	1
6	MULTIPOINT THERMOWELL	SS316	1
7	INLET DISTRIBUTOR	SS304	1
8	MANWAY 24 inch	SA-516-70	2
9	PSV NOZZLE N3	SS304	1
10	SKIRT SUPPORT	CARBON STEEL	1
11	GASKET (MANWAY)	SW SS316/GRAPHITE	2
12	CATALYST Pd/Al2O3	CATALYST	3.2 m3

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-3401, INTERLOCK SEQ-3401.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

This is sample data provided for CALIBER purposes only

SDK Polyolefin Expansion Project

TITLE: GENERAL ARRANGEMENT DRAWING

DWG No. TJC-LLD-GA-DC-3401A

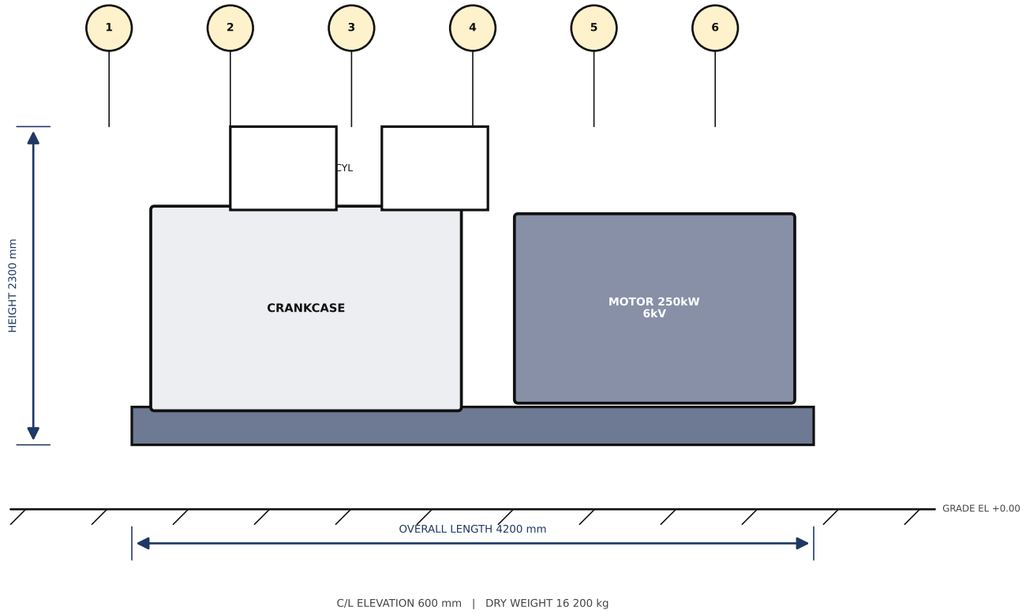
CATALYST REDUCTION REACTOR - Fixed-Bed Reactor (Pd Catalyst)

REV: 0

SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - RECYCLE GAS COMPRESSOR (KC-4501)

Type: Reciprocating Compressor (2-stage) | Area 4500 - GAS RECOVERY & COMPRESSION



C/L ELEVATION 600 mm | DRY WEIGHT 16 200 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	1ST STG SUCTION	6"-150#	RF	From KO drum FA-4510
N2	1ST STG DISCH	4"-300#	RF	To intercooler
N3	2ND STG SUCTION	4"-300#	RF	From intercooler
N4	2ND STG DISCH	3"-300#	RF	To reactor loop
N5	SEAL GAS	1"	SW	N2 buffer
N6	LUBE OIL	1"	SW	Force feed

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	CRANKCASE / FRAME	CAST IRON	1
2	CRANKSHAFT	FORGED STEEL	1
3	CONNECTING ROD	FORGED STEEL	2
4	CROSSHEAD	CAST IRON	2
5	PISTON (1ST STAGE)	ALUMINIUM	1
6	PISTON (2ND STAGE)	ALUMINIUM	1
7	PISTON RINGS	PTFE FILLED	8
8	SUCTION/DISCH VALVES	SS PLATE	8
9	PACKING SET	PTFE	2
10	MAIN BEARING	TRI-METAL	3
11	LUBE OIL PUMP	GEAR TYPE	1
12	INTERCOOLER EA-4502	SS TUBE	1

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-4501, INTERLOCK SEQ-4501.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

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SDK Polyolefin Expansion Project

TITLE:
GENERAL ARRANGEMENT DRAWING

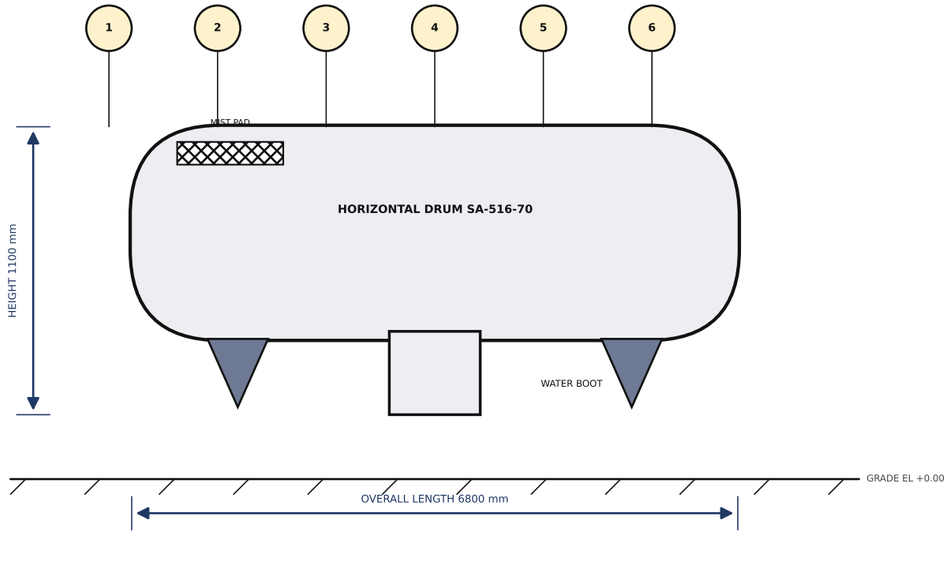
RECYCLE GAS COMPRESSOR - Reciprocating Compressor (2-stage)

DWG No.
TJC-LLD-GA-KC-4501

REV: 0
SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - SOLVENT HEATER (EA-5601)

Type: Shell & Tube Heat Exchanger (BEM) | Area 5600 - SOLVENT RECOVERY SYSTEM



C/L ELEVATION 550 mm | DRY WEIGHT 6 400 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	TUBE IN (SOLVENT)	6"-300#	RF	Cold hexane in
N2	TUBE OUT (SOLVENT)	6"-300#	RF	Hot hexane out
N3	SHELL IN (STEAM)	4"-150#	RF	MP steam in
N4	CONDENSATE OUT	3"-150#	RF	Condensate out
N5	SHELL VENT	1"	SW	Vent
N6	SHELL DRAIN	1"	SW	Drain

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	SHELL	SA-516-70	1
2	CHANNEL / BONNET	SA-516-70	1
3	TUBE BUNDLE	SS304L	1
4	TUBESHEET	SA-516-70 + SS304	2
5	BAFFLE PLATES	CARBON STEEL	9
6	TIE RODS & SPACERS	CARBON STEEL	6
7	CHANNEL COVER	SA-516-70	1
8	SHELL GASKET	SW SS316/GRAPHITE	2
9	CHANNEL GASKET	SW SS316/GRAPHITE	2
10	SADDLE SUPPORT	CARBON STEEL	2
11	VENT/DRAIN NOZZLE	SS304	4
12	STEAM TRAP ASSY	-	1

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-5601, INTERLOCK N/A (control loop only).

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

This is sample data provided for CALIBER purposes only

TITLE: GENERAL ARRANGEMENT DRAWING

SOLVENT HEATER - Shell & Tube Heat Exchanger (BEM)

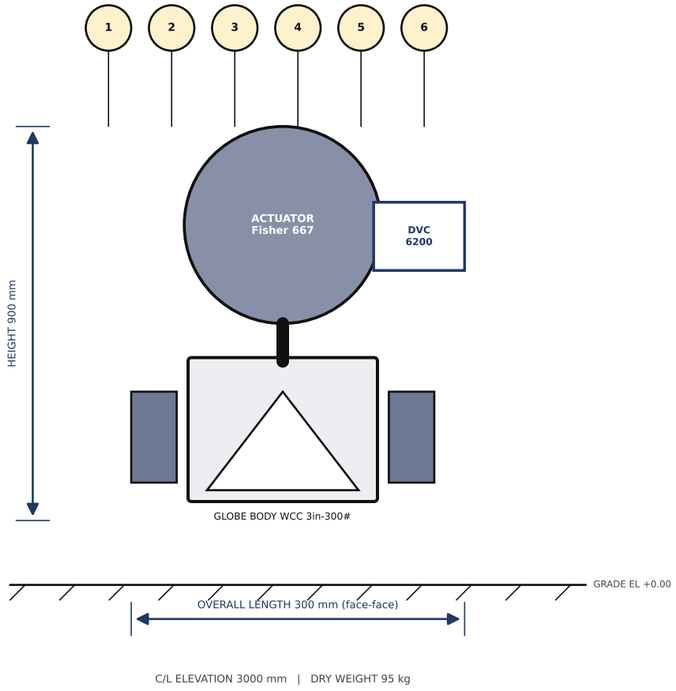
DWG No. TJC-LLD-GA-EA-5601

REV: 0

SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - SEPARATOR LEVEL CONTROL VALVE (LV-6701)

Type: Globe Control Valve + Diaphragm Actuator | Area 6700 - PRODUCT SEPARATION



C/L ELEVATION 3000 mm | DRY WEIGHT 95 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	INLET	3"-300#	RF	From FA-6710 bottoms
N2	OUTLET	3"-300#	RF	To FA-6720 flash

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	VALVE BODY	WCC	1
2	BONNET	WCC	1
3	CAGE	SS316	1
4	PLUG / STEM	SS316 + STELLITE	1
5	SEAT RING	SS316 + STELLITE	1
6	PACKING (PTFE V-RING)	PTFE	1 set
7	DIAPHRAGM	NBR/FABRIC	1
8	ACTUATOR SPRING	SPRING STEEL	1
9	POSITIONER DVC6200	-	1
10	BODY GASKET	SPIRAL WOUND	1
11	STEM CONNECTOR	STEEL	1
12	AIR FILTER REGULATOR	-	1

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-6701, INTERLOCK SEQ-6701.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

This is sample data provided for CALIBER purposes only

TITLE:
GENERAL ARRANGEMENT DRAWING

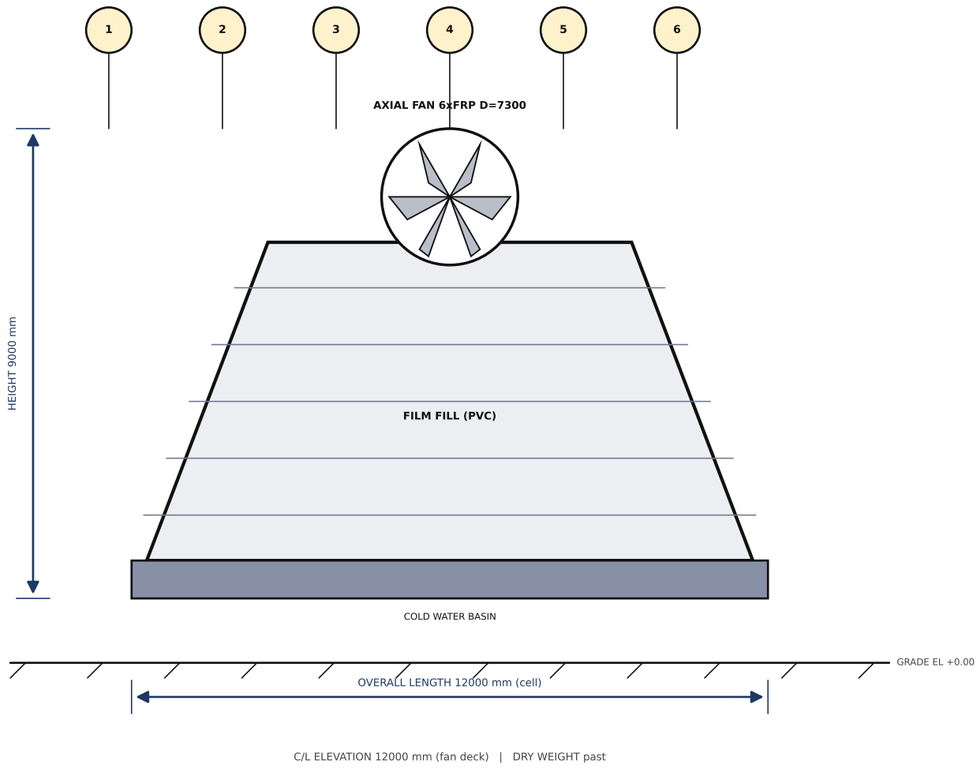
SEPARATOR LEVEL CONTROL VALVE - Globe Control Valve + Diaphragm Actuator

DWG No.
TJC-LLD-GA-LV-6701

REV: 0
SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - COOLING TOWER CELL FAN (CT-7801)

Type: Induced Draft Cooling Tower Fan | Area 7800 - COOLING WATER SYSTEM



C/L ELEVATION 12000 mm (fan deck) | DRY WEIGHT past

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	HOT WATER IN	24"	FLG	Warm return header
N2	COLD WATER OUT	30"	FLG	From cold basin
N3	MAKE-UP	6"	FLG	Make-up water
N4	BLOWDOWN	4"	FLG	Blowdown

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	FAN BLADE (FRP)	GRP	6
2	FAN HUB	CAST ALUMINIUM	1
3	GEARBOX (RIGHT ANGLE)	CAST IRON	1
4	FLOATING DRIVE SHAFT	COMPOSITE	1
5	MOTOR	TECO 55kW	1
6	FAN DECK / STACK	FRP	1
7	DRIFT ELIMINATOR	PVC	1 set
8	FILM FILL	PVC	1 set
9	GEARBOX OIL SEAL	NBR	2
10	FAN BEARING	SPHERICAL ROLLER	2
11	VIBRATION SWITCH	-	1
12	BARSCREEN	SS304	1

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-7801, INTERLOCK SEQ-7801.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

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TITLE:
GENERAL ARRANGEMENT DRAWING

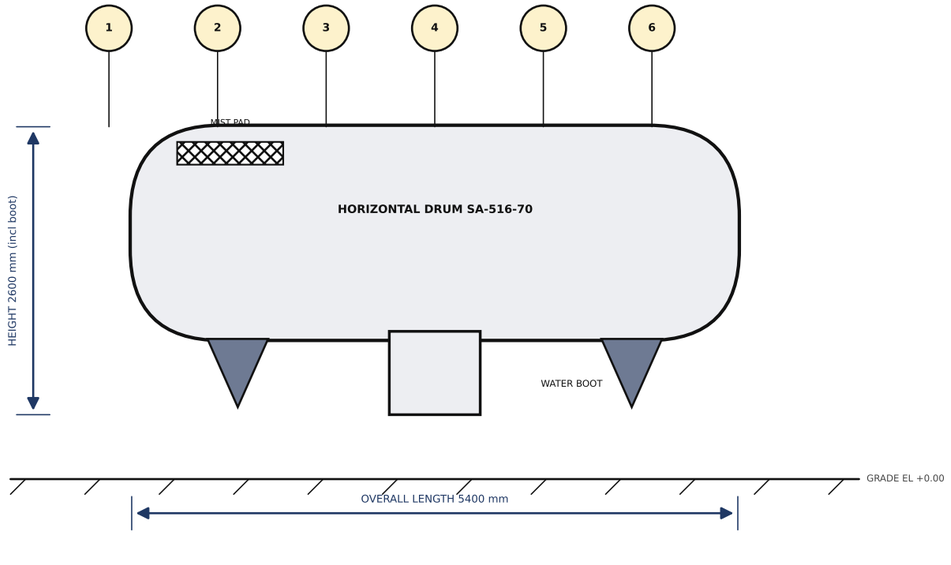
COOLING TOWER CELL FAN - Induced Draft Cooling Tower Fan

DWG No.
TJC-LLD-GA-CT-7801

REV: 0
SHEET 1 OF 1

GENERAL ARRANGEMENT DRAWING - REFLUX ACCUMULATOR DRUM (FA-8901)

Type: Horizontal Pressure Vessel | Area 8900 - SOLVENT FRACTIONATION



C/L ELEVATION 4500 mm | DRY WEIGHT 8 900 kg

NOZZLE / CONNECTION SCHEDULE

MARK	SERVICE	SIZE-RATING	FACE	REMARKS
N1	INLET	8"-150#	RF	Column overhead condensate
N2	REFLUX OUT	6"-150#	RF	To reflux pump
N3	PRODUCT OUT	4"-150#	RF	To product
N4	BOOT DRAIN	2"-150#	RF	Water draw-off
N5	PSV	3"-150#	RF	To PSV-8901
M1	MANWAY	24"	FLG	Access

BILL OF MATERIAL

No	PART NAME	MATERIAL	QTY
1	SHELL	SA-516-70	1
2	HEAD (2:1 ELLIP)	SA-516-70	2
3	WATER BOOT	SA-516-70	1
4	SADDLE SUPPORT	CARBON STEEL	2
5	MANWAY 24 inch	SA-516-70	1
6	VORTEX BREAKER	SS304	2
7	MIST PAD (INLET)	SS304 MESH	1
8	LEVEL BRIDLE	SA-106-B	1
9	PSV NOZZLE N5	SS304	1
10	NAMEPLATE & BRACKET	SS304	1
11	MANWAY GASKET	SW SS316/GRAPHITE	1
12	EARTHING BOSS	CARBON STEEL	2

NOTES:

1. MATERIAL OF CONSTRUCTION PER WPN-MAT-001.
2. SURFACE PREP & PAINTING PER WPN-PAINT-001.
3. ALL DIMENSIONS IN mm UNLESS NOTED.
4. WEIGHTS ARE ESTIMATED (DRY / ERECTION).
5. FOR NOZZLE LOADS REFER TO VENDOR DRAWING.
6. ASSOC DOCS: P&ID TJC-LLD-PID-8901, INTERLOCK SEQ-8901.

REVISION HISTORY

REV	DESCRIPTION	BY	DATE
A	ISSUED FOR REVIEW	RS	15-05-2026
B	ISSUED FOR APPROVAL	RS	27-05-2026
0	ISSUED FOR CONSTRUCTION	RS	05-06-2026

This is sample data provided
for CALIBER purposes only

TITLE:
GENERAL ARRANGEMENT DRAWING

REFLUX ACCUMULATOR DRUM - Horizontal Pressure Vessel

DWG No.
TJC-LLD-GA-FA-8901

REV: 0
SHEET 1 OF 1